

Reflection

Description

What is the app project?

My project was SourceSense, an Android education app designed to help university students improve their ability to evaluate evidence and academic sources. It contains Beginner, Intermediate, and Advanced learning modules covering skills such as identifying source types, checking research methods and samples, recognising bias, and judging the strength of evidence. Students answer short questions, receive immediate feedback and learning tips, track their progress, and search and review real academic sources through Crossref.

What ethical issues and ACS principles from Assessment 2 were relevant to the design?

The main ethical issues relevant to SourceSense were honesty, privacy, accessibility, user control, and avoiding misleading information. These principles encouraged me to make the app clear about its limitations, collect only necessary information, and support students without making decisions for them.

Feelings

What were my initial thoughts and reactions throughout the project?

At the beginning, I was excited about building SourceSense because the idea felt useful and different from a simple quiz app. However, I was also worried that the project might become too large, especially when I added different difficulty levels, progress tracking, real source searching, and source review features.

During development, I often felt frustrated when the app did not behave as expected. Some Crossref results had missing information, PDF links did not always open properly, and several UI tests failed because some page content had to be scrolled into view first. I also redesigned some screens several times because the early versions looked too crowded or included information that was not very useful.

At the same time, I became more confident whenever I solved a problem. When the learning modules worked, the statistics updated correctly, real-source reviews could be saved, and all tests passed, I felt proud of the final result.

How did I feel when applying ethical principles to technical decisions?

At first, I was unsure how ideas such as honesty and professional responsibility could be applied to actual app features. Later, when I was building the Crossref feature, I realised

that students might think every paper in the search results was reliable. This made me more careful about the wording and the way the source review page was designed. It also made me realise that, as the developer, I was responsible for how users understood the information shown in the app.

Evaluation

[What went well during the development process?](#)

Several parts of the project worked well. The learning process was clear, from choosing a module and answering questions to receiving feedback and viewing the final result. The Crossref search also made the app more practical because students could move from practice cases to real academic sources. I was also satisfied with the Statistics and Settings pages, as students could track their progress and adjust their learning experience. The main features, including saving results and source reviews, worked as intended in the final version.

[What did not go well during the development process?](#)

The development process was not always smooth. Some early screens were too crowded, and some statistics did not add much value, so I later removed or simplified them. Crossref results were sometimes incomplete, and publisher links did not always provide a readable PDF. I also spent more time than expected fixing navigation, page layouts, and UI tests.

[What went well and what did not when applying ethical considerations?](#)

One thing that worked well was the design of the Crossref review feature. The app shows publication details, but it still asks students to check the source's relevance, method, and limitations themselves. This avoids giving students the idea that every paper found through Crossref is automatically suitable for their work.

However, I did not consider every issue at the beginning. For example, I originally added several detailed review statistics, but they only counted the options selected by students and did not clearly show whether their evaluation skills had improved. I later removed these statistics and kept only the total number of completed reviews. This made the Statistics page simpler and prevented the app from presenting weak data as meaningful learning results.

Analysis

[Why did the main design decisions work?](#)

The learning structure was one of the strongest parts of SourceSense. Beginner,

Intermediate, and Advanced modules allowed students to practise different evidence-evaluation skills step by step. Feedback after each question explained why an answer was correct or incorrect, while the Statistics page showed weaker areas and suggested what to practise next. Crossref then allowed students to use these skills with real academic sources instead of only fictional examples.

How did ethical issues and Android features shape the app?

SourceSense does not require an account, and learning records are stored locally through Room and DataStore. Users can also remove their saved history. The app only uses the permissions needed for Crossref searches and publication links. Larger text, reduced animations, and sound feedback settings also allow students to adjust how they use the app.

Android features helped connect the different parts of the project. ViewModels managed screen states, Room stored results and reviews, DataStore saved settings, and Compose Navigation linked the learning, search, statistics, and settings pages. Automated tests also helped me check saving, feedback, buttons, and error states.

Why did some decisions fail?

One early problem was that changing the difficulty level did not make much difference to the actual learning experience. The app showed Beginner, Intermediate, and Advanced options, but the number and type of questions were still too similar. I later reorganised the content into separate topic modules for each level, so students could practise skills that matched their selected difficulty. This showed me that a setting is only useful when it creates a clear change for the user.

Conclusion

What did I learn about ethical practice?

I learned that an education app should not make claims that its data cannot support. For SourceSense, Crossref could provide publication details, but students still needed to examine the source and make their own judgement.

What did I learn about professional practice?

I learned that completing a feature does not mean it is ready. I still needed to test it, fix unexpected problems, and change the design when something was unclear or did not work well.

What did I learn about technical practice?

I became more confident in connecting Compose screens, ViewModels, navigation,

Crossref, Room, and DataStore. Seeing all these parts work together in the final app gave me a real sense of achievement.

Action Plan

[What would I change next time?](#)

In future projects, I would plan the main features and testing process earlier so that major problems could be found before the app became too large.

[How would I apply these lessons in future projects?](#)

In the next version, I would add a source comparison feature so students could examine two or three papers side by side and compare their methods, samples, findings, and limitations. I would also expand the question bank and adjust later questions based on the areas where a student often makes mistakes, making practice more focused and less repetitive. Finally, I would ask real students to test these features and use their feedback to improve the difficulty, explanations, and overall learning flow.